SLOPO GUI

User Guide

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Contents

1 Getting Started

1.1 Pre requisites

- IBM PC with standard COM port cable
- Standard serial cable
- CD-ROM Drive.
- OS: MS Windows XP, Windows 7
- The .NET Framework version 2.0 re-distributable package. This is a free application available for download from Microsoft. http://www.microsoft.com/downloads
- Windows languages supported: US, Canada, South America, Japan, China, Korea, Thailand and India. France and Germany are currently not supported due to conflicts with the usages of commas and periods.

1.2 Installation Procedure

- Make sure that .NET Framework version 2.0(or upper) is installed. This is available from Microsoft as a free download.
- Insert the CD into the CD-ROM drive. For this example, assume that it is the D: drive.
- Before starting, it is recommended that you close any open applications.
- Open Windows Explorer by Right-clicking the Start button and clicking Explore, or any other technique that you are familiar with.
- Double click on the D: drive
- Double click on the D:\Install directory
- Double click on SETUP.EXE
- Follow the instructions in the setup program.

Remove the CD-ROM and store it in a safe place. The CD is not needed to run the software after the installation is complete.

1.3 Setting up the communication Parameters

1.3.1 Specifying the Connected Port for the First Time

Once the SLOPO started for the first time, user can specify the serial port, to which the SLOPO hardware is connected. Please specify the correct port number in the following form.

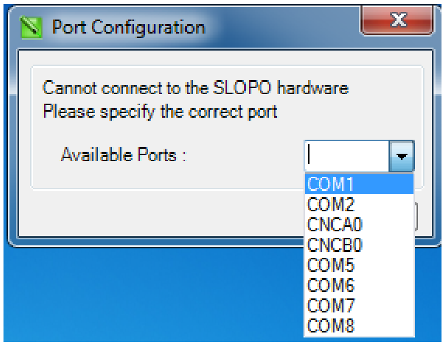

Figure 1.3.1.1: Port Selection at first time

1.3.2 Specifying the Connected Port Later

If this port needs to changed afterwards, please go to

Tools -> Settings -> Slopo submenu to launch Options dialog box.

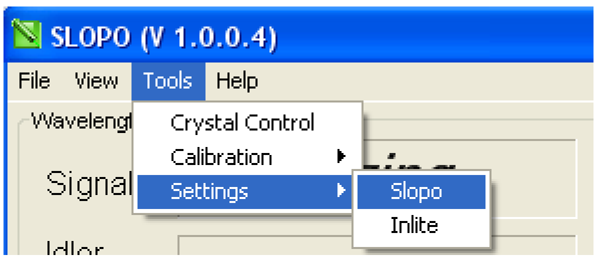

Figure 1.3.2.1: Getting SLOPO settings

Specify the port number where the Slopo hardware is connected. It is recommended to keep the default baud rate 9600.

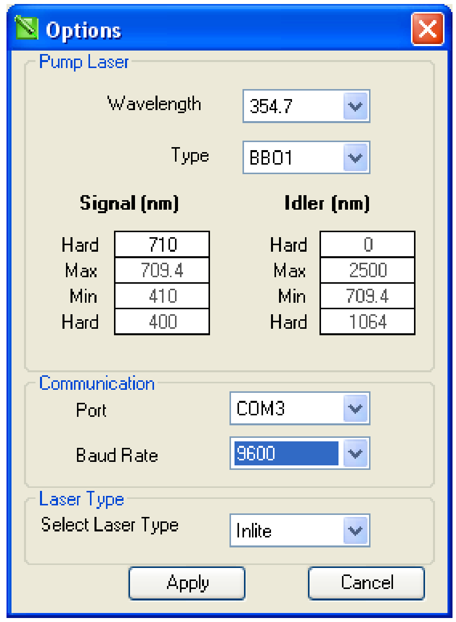

Figure 1.3.2.1: Selecting SLOPO Port

1.4 Launching the SLOPO GUI

Once the SLOPO GUI is successfully installed and the SLOPO hardware is connected, launch the GUI by

1. Double clicking on SLOPO GUI icon on desktop OR
2. Go to Start -> Program Menu -> Continuum -> select Slopo short cut icon

During the initialization process, GUI prompt user to make sure that the output shutter of the pump laser is closed. This is to avoid any damages to the crystal.

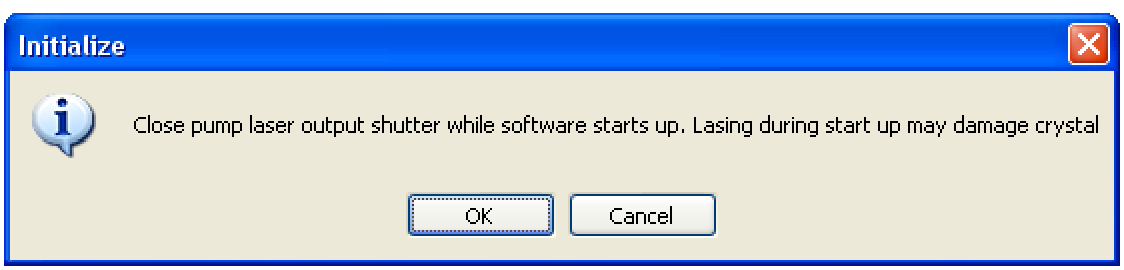

Figure 1.4.1: Warning user about the Laser shutter

Once SLOPO GUI is started, you can modify the communication port if necessary as explain in the above chapter.

2 Main GUI

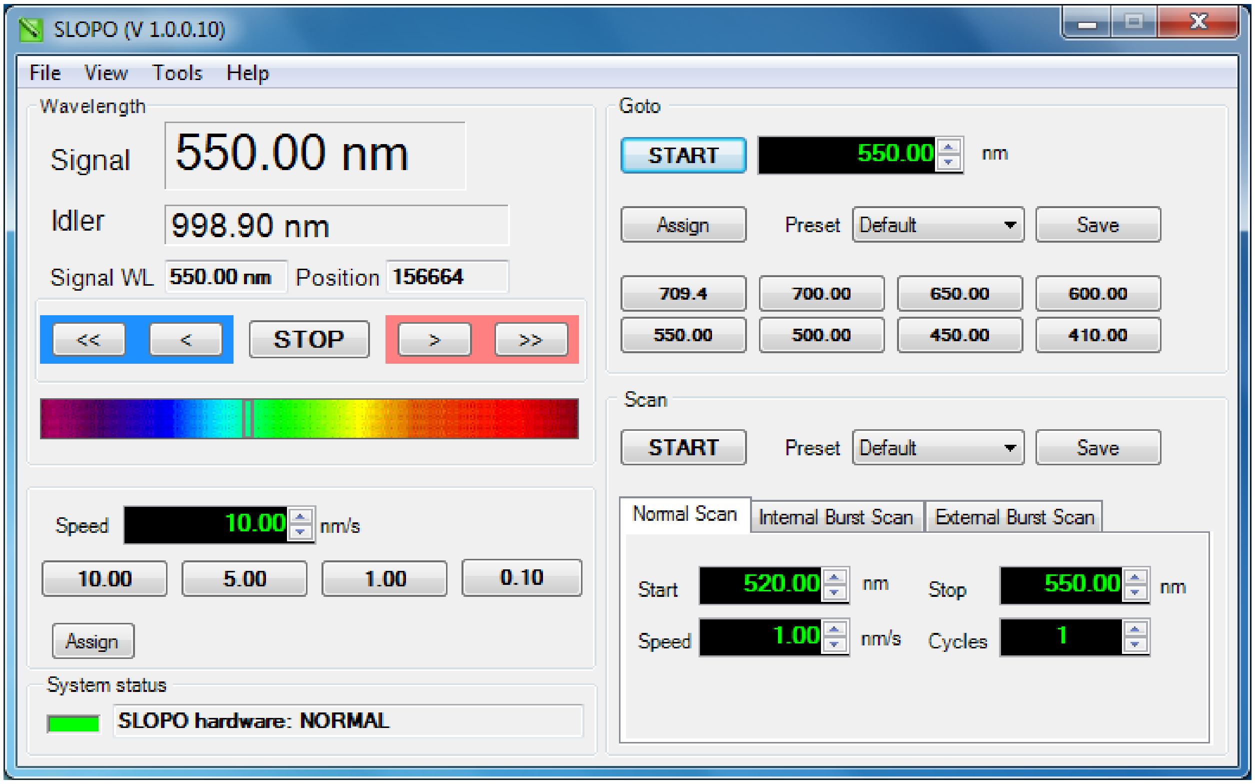

Figure 2.1: SLOPO GUI main Screen

SLOPO Main GUI consists of all the necessary features for the end users, so they can to have the full control over the system to achieve the desired output. This main form is consists of many sub controls, which are specific to different operations.

The main GUI components include;

1. Wavelength Display panel
2. Go To Operation panel
3. Scan Operation panel
4. System Status Panel
5. Speed Control Panel
6. Menu bar

Functionality and the operations of these panels will be discussed in detail under following chapters.

3 Functionalities and Operations

3.1 Wavelength Display

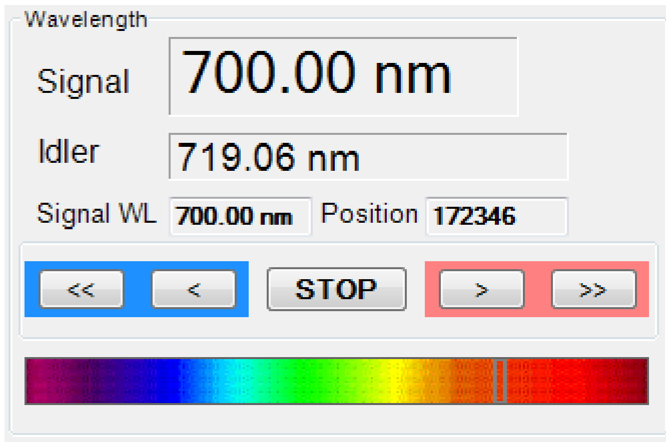

Figure 3.1.1: Wavelength Display Panel

This panel displays the real time system Signal and Idler values. In addition, user can perform following operations.

1. Step Up or Step Down current wavelength by predefine value.
   By pressing > or < buttons, user can increase or decrease the wavelength by predefine value. Wavelength will keep on increase or decrease as far as the > or < buttons are been pressed.

2. Continuously Scan the wavelength forward or backward
   By pressing >> or << buttons, user can continuously scan the wavelength upwards or downwards from the current value. This continuous operation can be terminated by pressing the "STOP" button.

3.2 Go To Operation

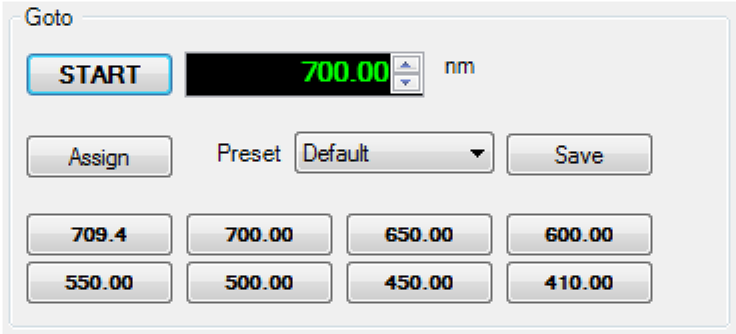

Figure 3.2.1: Go To Operation Panel

Go To Panel supports to move to a specific wavelength by just a one click. There are 8 predefined wavelengths in display and there are 3 preset groups to select and assign.

Once the Destination wavelength is selected (or typed), press start button to initiate the "Go To" Process. Crystal (motor) will start moving with the speed, which is specified in Speed Control Panel.

During the crystal (Motor Movement), the text of the start button becomes "STOP". So if it is required to stop the movement, please press the STOP button.

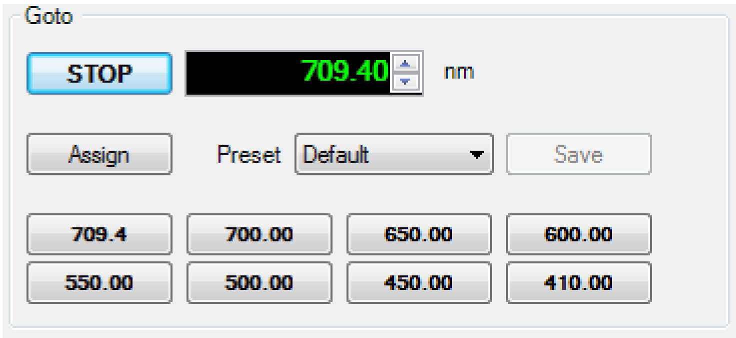

Figure 3.2.1: Go To Operation Panel during the Crystal movement

3.2.1 Assigning Preset Values

User can assign preset values as he prefers.

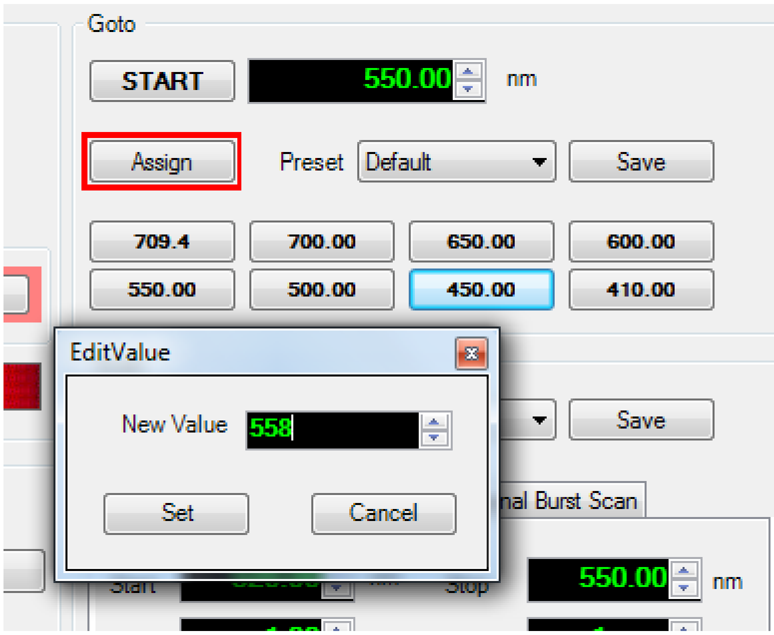

Figure 3.2.1.1: Assigning Preset Values

Follow the below procedure for that;

1. Click on Assign Button. It will be marked in red color.
2. Then click on the preset button, to which the new value going to be assigned. It is 450 in the above case.
3. Edit Value dialog will appear and insert the new value and say "Set". New value will be assigned to the preset button.

Note: Similar procedure can be followed to assign preset values for the other panels.

3.3 Scan Operation Panel

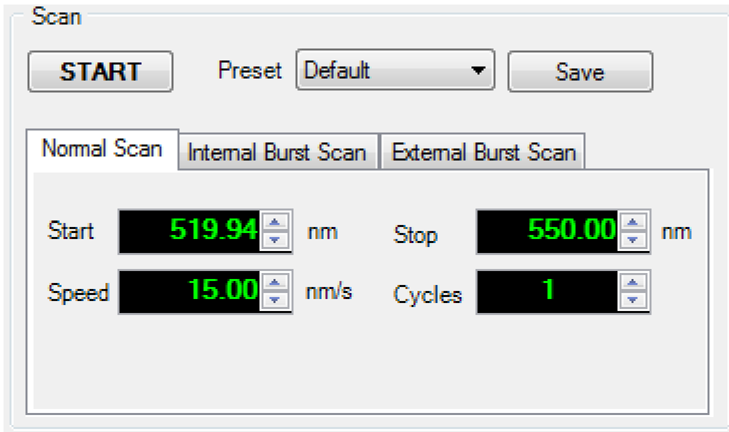

Figure 3.3.1: Scan Operational Panel

Scanning panel support scan from one specific (start) wavelength to another wavelength (stop) with different options. Scanning can be done from smaller wavelength value to a higher wavelength or other way.

Scanning feature has 3 main categories.

1. Normal Scan
2. Internal Burst Scan
3. External Burst Scan

Once the required option selected and parameters specified, press the “START” button to initiate the scan operation.

3.3.1 Normal Scan Operation

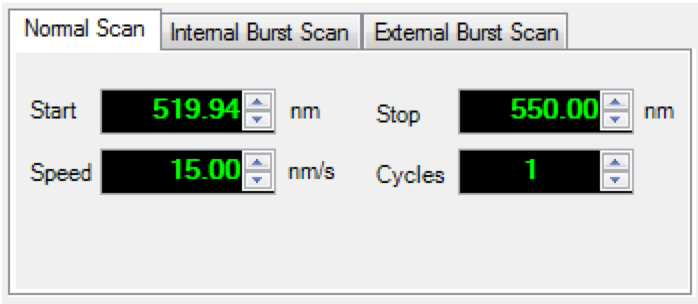

Figure 3.3.1.1: Normal Scan Operational Panel

Start: From which wavelength scanning should start

Stop: Up to what wavelength scanning should be performed

Cycles: How many scanning cycles needs to perform

Speed: Scanning speed nm per second

In the normal scan procedure, Crystal will move from the start wavelength to end wavelength with the specified speed.

During the reverse scanning and first time moving the starting wavelength (just before the scanning cycle starts), system will use the speed, which is specified in the speed control panel.

If it's required to abort the scanning operation, press the "STOP" button on the wavelength panel.

3.3.2 Internal Burst Scan Operation

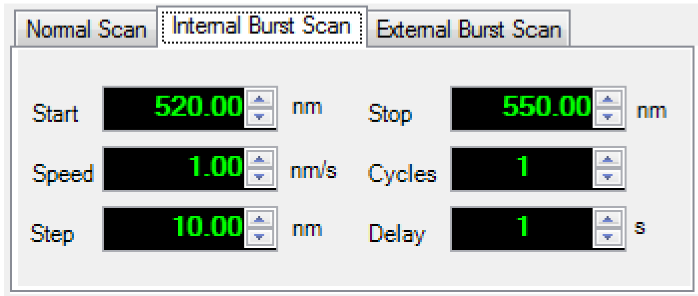

Figure 3.3.2.1: Internal Burst Scan Operational Panel

Start: From which wavelength scanning should start

Stop: Up to what wavelength scanning should be performed

Cycles: How many scanning cycles needs to perform

Speed: Scanning speed nm per second

Step: Scanning step in nm.

E.g. In the above example, since the step is 10, scanning will happen from 510,530,540 and 550.

Delay: How many seconds the system delay once it reaches to the next scanning step

In the Internal Burst scan procedure, Crystal will move from the start wavelength to end wavelength with the specified speed. Once it advances by the specified scan step, system will halt for the number of seconds, specified in the Delay value.

During the reverse scanning and first time moving the starting wavelength, system will use the speed, which is specified in the speed control panel. Please note that the step value is not followed during the reverse operation.

If it's required to abort the scanning operation, press the "STOP" button on the wavelength panel.

3.3.3 External Burst Scan Operation

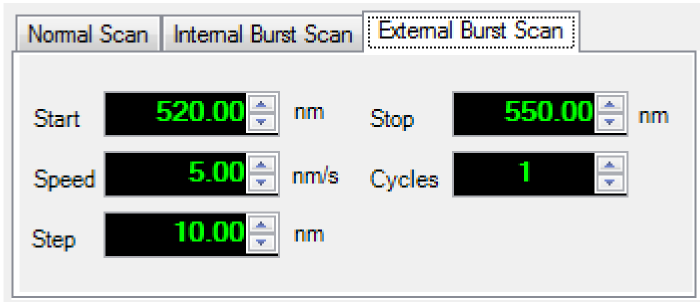

Figure 3.3.3.1: External Burst Scan Operational Panel

Start: From which wavelength, scanning should start

Stop: Up to what wavelength, scanning should be performed

Cycles: How many scanning cycles needs to perform

Speed: Scanning speed nm per second

Step: Scanning step in nm.

In the External Burst scan procedure, Crystal will move from the start wavelength to end wavelength with the specified speed. Once the motor moves to by specific steps, it will stop and wait for user to press the "continue" button.

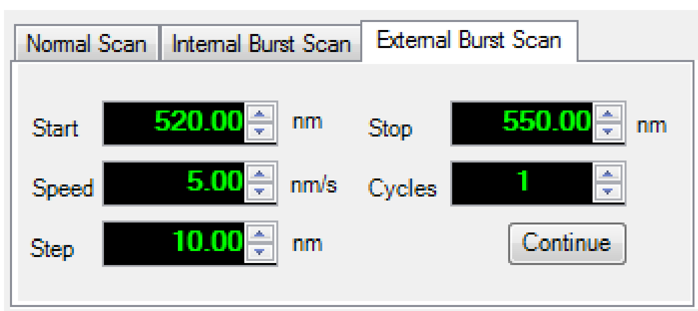

Figure 3.3.1.2: Normal Scan Operational Panel when waiting for the user input

3.4 Setting up System Speed

Figure 3.4.1: setting up the speed for the crystal movements

System Scan speed can be specify from the available predefined options of 0.10, 1.00, 5.00 & 10.00 nm per second or change by moving up or down arrow buttons.

3.5 System Status

System Status display the Errors & Warnings related to the SLOPO system and the connectivity.

During the normal operation, this will display that the system operates normally.

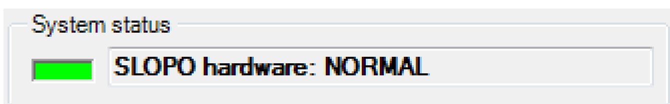

Figure 3.5.1: System status display during the normal operation

If in error condition, this will display any error associate with the system.

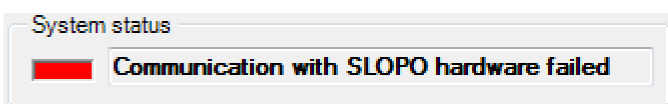

Figure 3.5.2: System status display when communication failed with the hardware

3.6 Other GUI Operations

3.6.1 Full Mode or Skin Mode

By clicking on Full Mode or Skin Mode menu items, user can switch between normal full mode operation and compact mode operation.

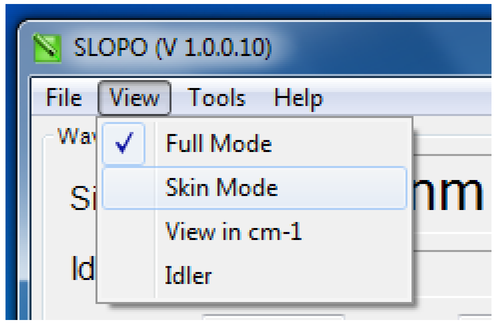

Figure 3.6.1: Selecting GUI appearance

In the compact mode operation, application will look like below.

Figure 3.6.2: GUI in compact mode

3.6.2 View in nm or cm-1

By default, application will display wavelengths in nm. To switch between nm and cm-1, select menu items “View in nm” or “View in cm-1”.

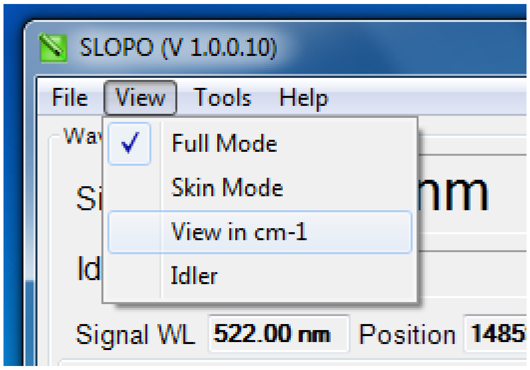

Figure 3.6.2.1: Switching between nm or cm-1 mode

Below picture illustrates when the GUI in cm-1 mode.

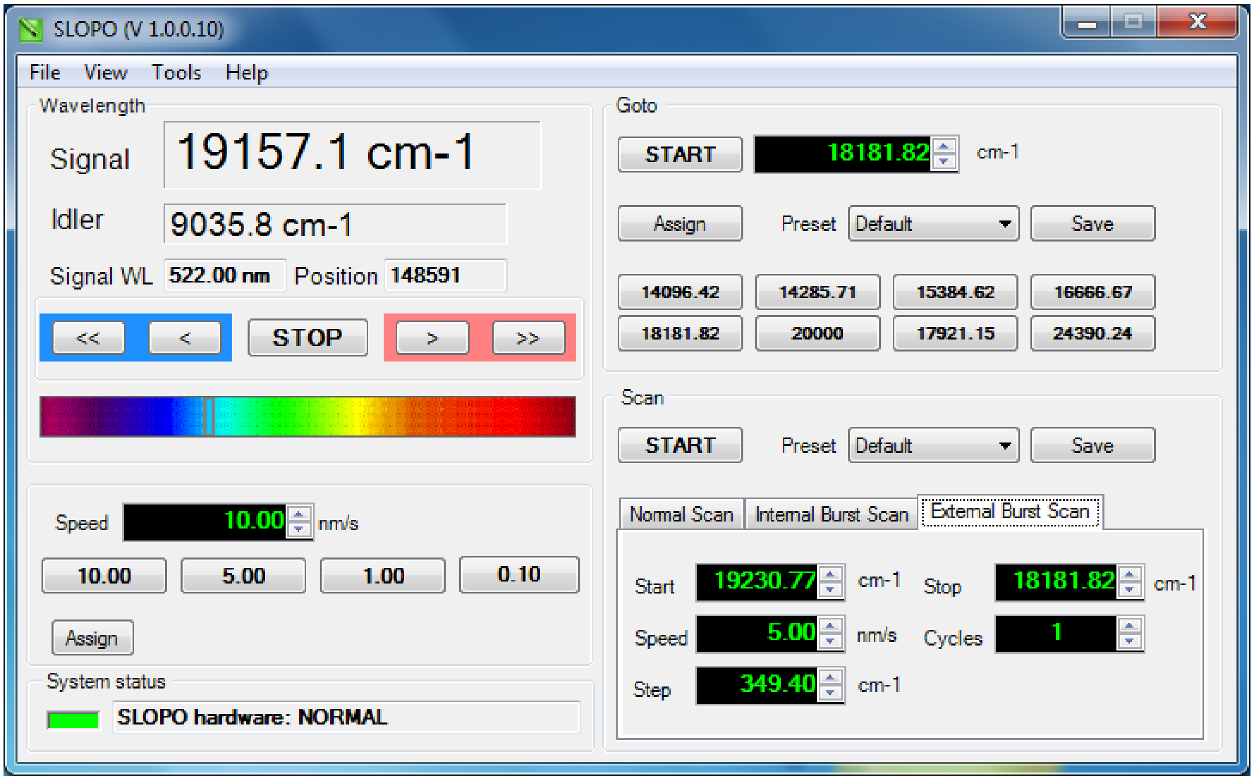

Figure 3.6.2.2: GUI in cm-1 mode

4 Calibration

SLOPO GUI provides facilities to edit existing calibration data or create new calibration files.

Go to Tools menu then Calibration Submenu and then select Crystal option.

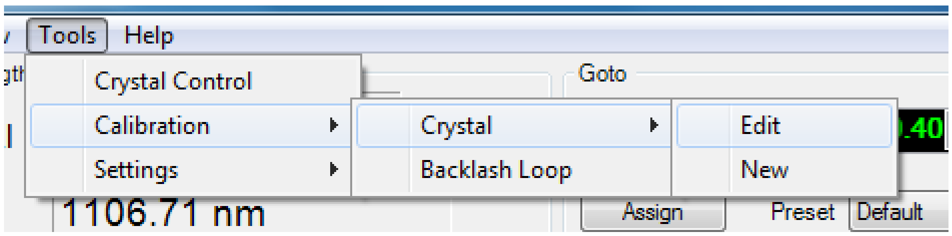

Figure 4.1: Getting the calibration window

Warning:

It is recommended that the calibration should be handled by a trained person who has the relevant knowledge and the authority over the system.

4.1 Calibration Curve

Graph Visualizer window can be achieved by clicking on “Show Graph” button in Edit/New Crystal Calibration window.

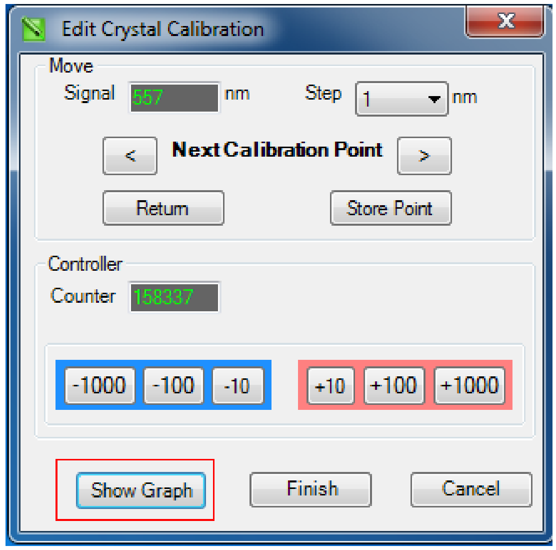

Figure 4.1.1: How to get the Graph Visualizer

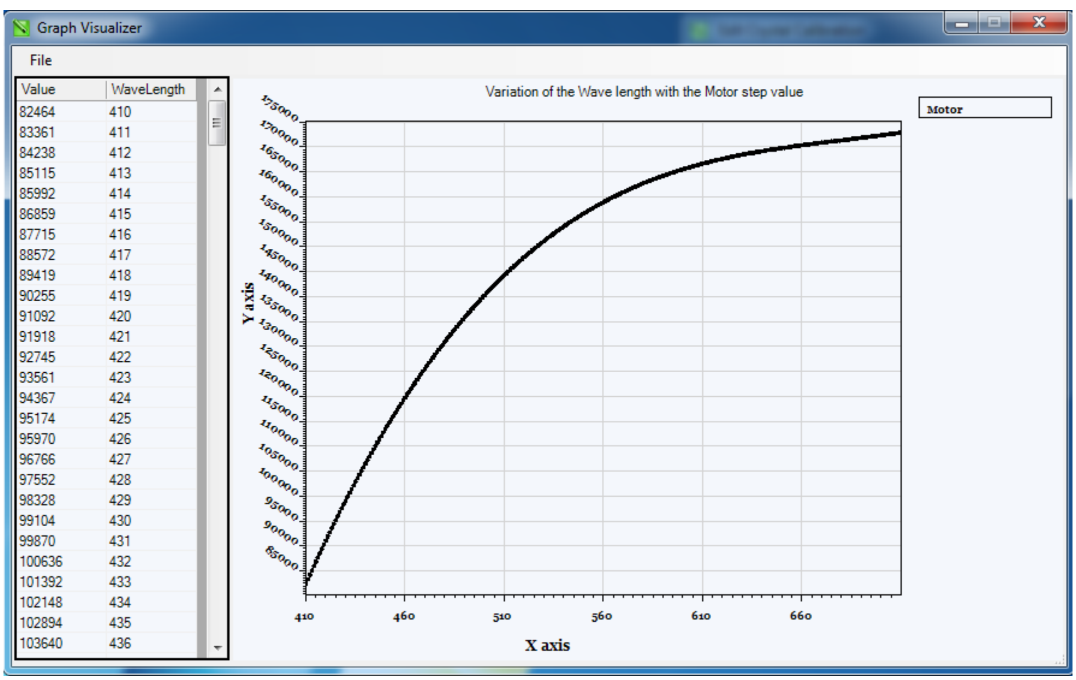

Figure 4.1.2: Calibration Curve

Graph Visualizer Form displays the wavelength to motor position mapping table and the curve.

4.2 Edit Existing Calibration

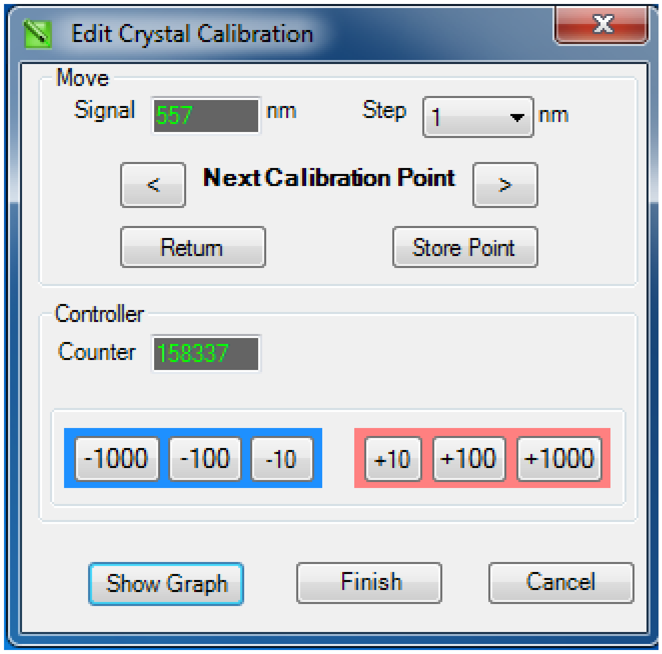

Figure 4.2.1: Editing the Crystal Calibration

Edit Crystal Calibration form enables to change the wavelength to motor position map by moving the motor by predefine steps forward or backwards.

This form will display the current system wavelength and the relevant motor counter position. User can change this counter position by step value of 10,100 or 1000 in up or down direction by pressing the relevant button.

Once required changes were done to the particular wavelength to counter mapping, press the "Store Point" button to store the new counter value in the application memory.

Likewise, user can change next wavelength and so on. Once all the necessary changes for the calibration file were done, press the Finish button to update loaded calibration file with the changes.

Show graph will display the modified calibration curve with the wavelength values and the motor counter positions.

4.3 Create a New calibration file

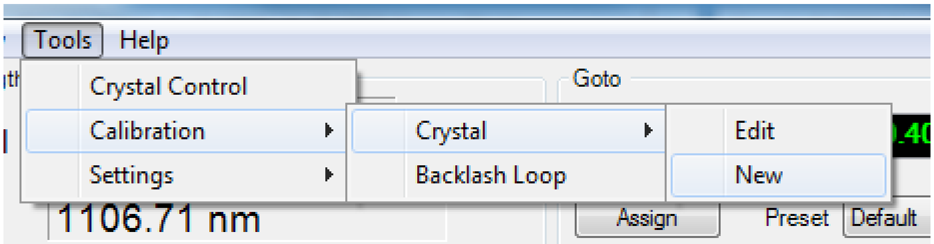

Figure 4.3.1: Getting the calibration window

User can create a new calibration file by selecting the "New" child menu under Crystal submenu. This operations are similar to the edit calibration process, but here user will be able to create a new file.

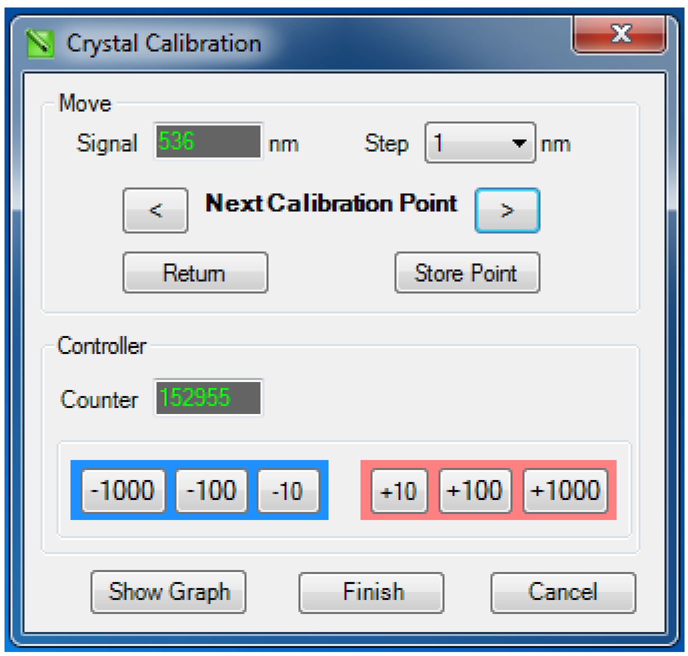

Figure 4.3.2: Creating new calibration file

Once you complete the calibration, press the Finish button and it will prompt save the calibration file.

4.4 Loading an Existing Calibration File

User can load a new calibration file by clicking on "Load New Data File" submenu available under File menu.

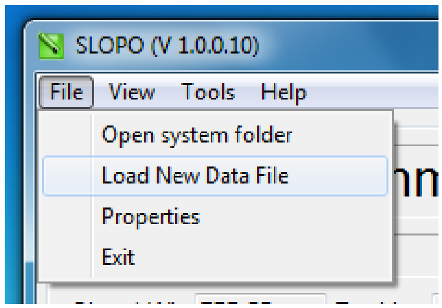

Figure 4.4.1: Loading new calibration file

This will pop up a file open dialog box with list of calibration files available under Calibration File folder.

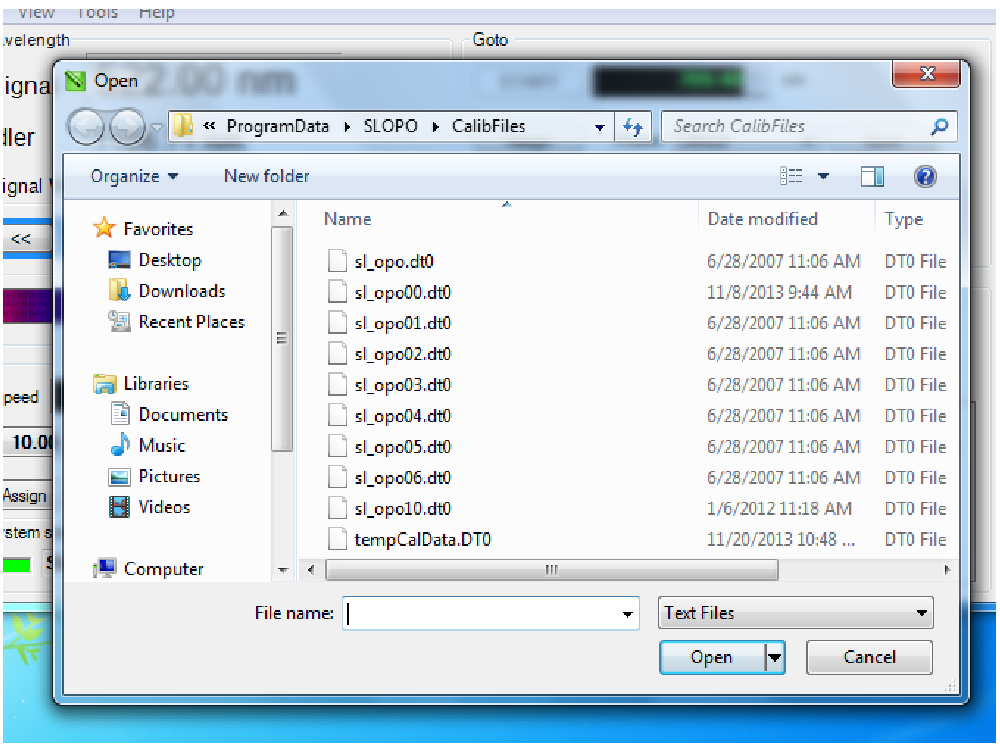

Figure 4.4.2: Selecting a calibration file

4.5 View Current Calibration File Properties

Current Calibration File properties such as file name, Minimum and Maximum wavelength and respective counter values can be view by clicking Properties Submenu under File Menu.

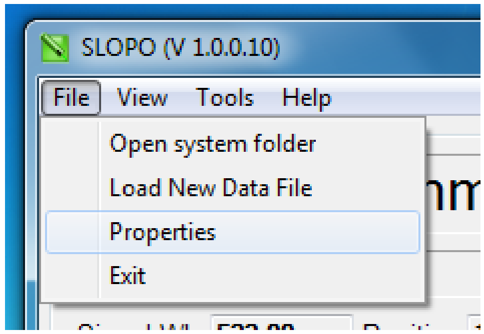

Figure 4.5.1: How to reach the properties window

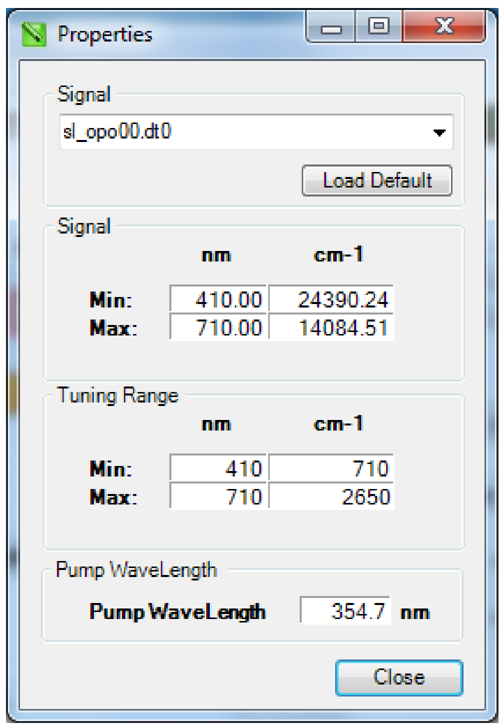

Figure 4.5.2: Loaded calibration file properties

4.6 Compensating the Backlash

Use Backlash window provide enter the value to compensate motor backlash.

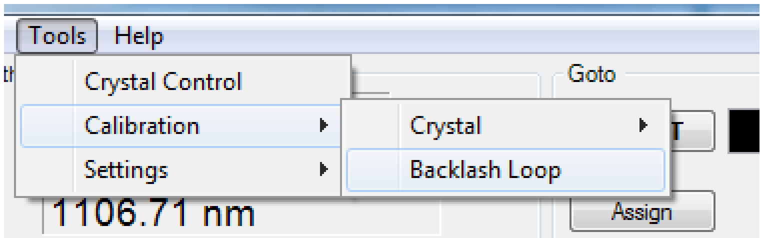

Figure 4.6.1: How to reach the Backlash Window

Follow the above steps to achieve the Backlash window.

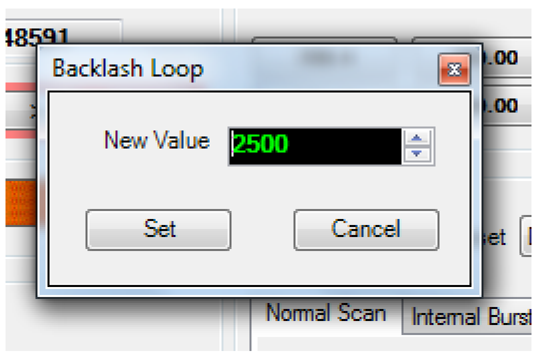

Figure 4.6.2: Specify the backlash steps

System will use the above inserted value to compensate the motor backlash during the operation.